

Project Synopsis Report

Project Title: Smart Asset Management System

1. Introduction

Organizations allocate various assets such as laptops, desktops, monitors, mobile devices, ID cards, software licenses, and other equipment to employees. Managing these assets manually often leads to inaccurate inventory, misplaced assets, delayed returns, and poor accountability.

The Smart Asset Management System is a web-based application developed using Spring Boot that automates the complete lifecycle of organizational assets. It enables administrators to add, assign, track, maintain, and recover assets while maintaining a complete history of every transaction. Employees can view assigned assets, submit return requests, and report damaged or lost assets.

2. Problem Statement

Manual asset management using spreadsheets or paper records results in poor tracking, data duplication, delayed returns, limited reporting, and reduced accountability. A centralized digital solution is required.

3. Objectives

- Develop a centralized asset management platform.
- Maintain complete asset records.
- Automate assignment and return workflows.
- Implement secure role-based authentication.
- Generate reports and analytics.
- Maintain audit logs.

4. Proposed System

The system provides separate dashboards for administrators and employees. Administrators can manage employees, departments, assets, vendors, assignments, maintenance, and reports. Employees can securely log in, view assigned assets, submit return requests, and report damaged or lost assets.

5. Major Modules

- Authentication & Authorization (JWT)
- Employee Management
- Asset Management
- Asset Assignment & Return
- Maintenance Management
- Reports & Dashboard
- Notifications

6. Functional Requirements

User login, employee management, department management, asset management, assignment/return, search & filters, reporting, dashboard analytics, notifications, audit logs.

7. Non-Functional Requirements

Security, scalability, performance, usability, maintainability, reliability, and data integrity.

8. Technology Stack

Frontend: HTML5, CSS3, Bootstrap, JavaScript, Thymeleaf

Backend: Java 21, Spring Boot, Spring MVC, Spring Data JPA, Spring Security, JWT

Database: H2

Build Tool: Maven

API Testing: Postman

Documentation: Swagger/OpenAPI

Version Control: Git & GitHub

9. Proposed Database Tables

Users, Roles, Employees, Departments, Assets, Asset Categories, Asset Assignments, Vendors, Maintenance Records, Notifications, Audit Logs.

10. Expected Outcomes

A centralized platform that improves asset visibility, accountability, inventory management, maintenance tracking, and reporting while reducing manual effort.

11. Conclusion

The Smart Asset Management System streamlines organizational asset tracking through automation, secure authentication, centralized data management, and real-time reporting.

The solution is scalable and can be extended with cloud technologies and smart tracking capabilities.